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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.2.3
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
## Warning: package 'tibble' was built under R version 4.2.3
```

```
## Warning: package 'tidyr' was built under R version 4.2.3
```

```
## Warning: package 'readr' was built under R version 4.2.3
```

```
## Warning: package 'dplyr' was built under R version 4.2.3
```

```
## Warning: package 'forcats' was built under R version 4.2.3
```

```
## Warning: package 'lubridate' was built under R version 4.2.3
```

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```
library(openintro)
```

```
## Warning: package 'openintro' was built under R version 4.2.3
```

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```
library(statsr)
```

```
## Warning: package 'statsr' was built under R version 4.2.3
```

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```
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.2.3
```

Exercise 1

What are the dimensions of the dataset? What does each row represent?

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```
dim(hfi)
```

```
## [1] 1458 123
```

[Hide](#)

```
tibble(hfi)
```

```
## # A tibble: 1,458 × 123
##   year ISO_code countries region pf_rol_procedural pf_rol_civil
##   <dbl> <chr>   <chr>   <chr>         <dbl>     <dbl>
## 1  2016 ALB     Albania Eastern Europe      6.66      4.55
## 2  2016 DZA     Algeria Middle East & North... NA        NA
## 3  2016 AGO     Angola  Sub-Saharan Africa  NA        NA
## 4  2016 ARG     Argentina Latin America & the... 7.10      5.79
## 5  2016 ARM     Armenia  Caucasus & Central ... NA        NA
## 6  2016 AUS     Australia Oceania            8.44      7.53
## 7  2016 AUT     Austria  Western Europe      8.97      7.87
## 8  2016 AZE     Azerbaijan Caucasus & Central ... NA        NA
## 9  2016 BHS     Bahamas Latin America & the... 6.93      6.01
## 10 2016 BHR     Bahrain  Middle East & North... NA        NA
## # i 1,448 more rows
## # i 117 more variables: pf_rol_criminal <dbl>, pf_rol <dbl>,
## #   pf_ss_homicide <dbl>, pf_ss_disappearances_disap <dbl>,
## #   pf_ss_disappearances_violent <dbl>, pf_ss_disappearances_organized <dbl>,
## #   pf_ss_disappearances_fatalities <dbl>, pf_ss_disappearances_injuries <dbl>,
## #   pf_ss_disappearances <dbl>, pf_ss_women_fgm <dbl>,
## #   pf_ss_women_missing <dbl>, pf_ss_women_inheritance_widows <dbl>, ...
```

Exercise 2

The dataset spans a lot of years, but we are only interested in data from year 2016. Filter the data hfi data frame for year 2016, select the six variables, and assign the result to a data frame named hfi_2016.

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```
hfi_2016 <- hfi %>%
  filter(year == 2016) %>%
  select(year, hf_score, pf_score, pf_expression_control)
hfi_2016
```

```
## # A tibble: 162 × 4
##   year hf_score pf_score pf_expression_control
##   <dbl>   <dbl>   <dbl>         <dbl>
## 1  2016    7.57    7.60          5.25
## 2  2016    5.14    5.28           4
## 3  2016    5.64    6.11          2.5
## 4  2016    6.47    8.10          5.5
## 5  2016    7.24    6.91          4.25
## 6  2016    8.58    9.18          7.75
## 7  2016    8.41    9.25           8
## 8  2016    6.08    5.68          0.25
## 9  2016    7.40    7.45          7.25
## 10 2016    6.85    6.14          0.75
## # i 152 more rows
```

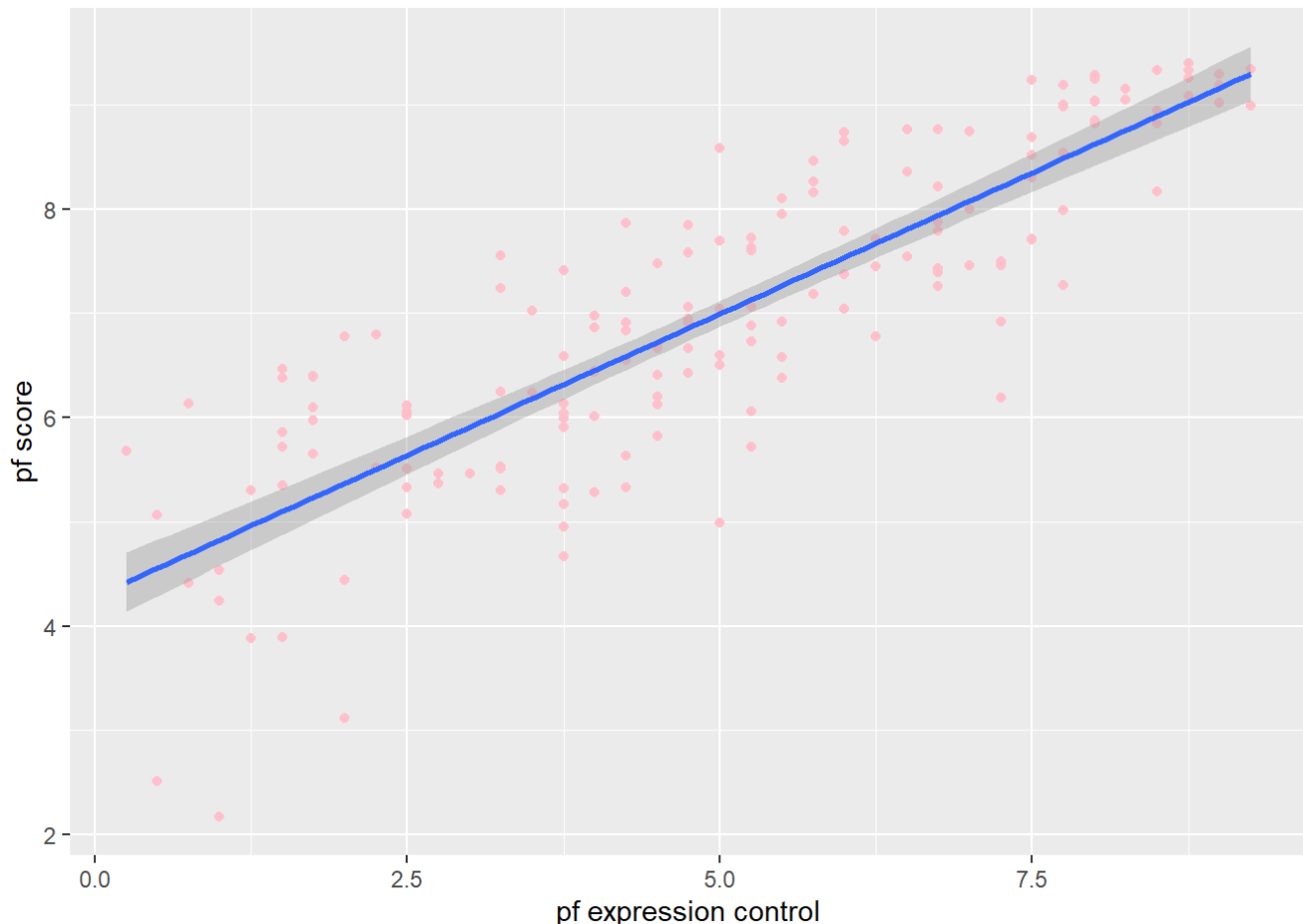
Exercise 3

What type of plot would you use to display the relationship between the personal freedom score, `pf_score`, and `pf_expression_control`? Plot this relationship using the variable `pf_expression_control` as the predictor. Does the relationship look linear? If you knew a country's `pf_expression_control`, or its score out of 10, with 0 being the most, of political pressures and controls on media content, would you be comfortable using a linear model to predict the personal freedom score?

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```
ggplot(hfi_2016, aes(pf_expression_control, pf_score)) +  
  geom_point(color = "pink") +  
  geom_smooth(method = 'lm', se = TRUE) +  
  xlab("pf expression control") +  
  ylab ("pf score")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Scatterplot because it is easy to see and allows us to see the relationship between the two variables. The relationship looks linear and some points are scattered. At first glance, I would not feel comfortable using this to predict a linear model.

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```
hfi_2016 %>%  
summarise(cor(pf_expression_control, pf_score))
```

```
## # A tibble: 1 × 1  
##   `cor(pf_expression_control, pf_score)`  
##                                     <dbl>  
## 1                                     0.845
```

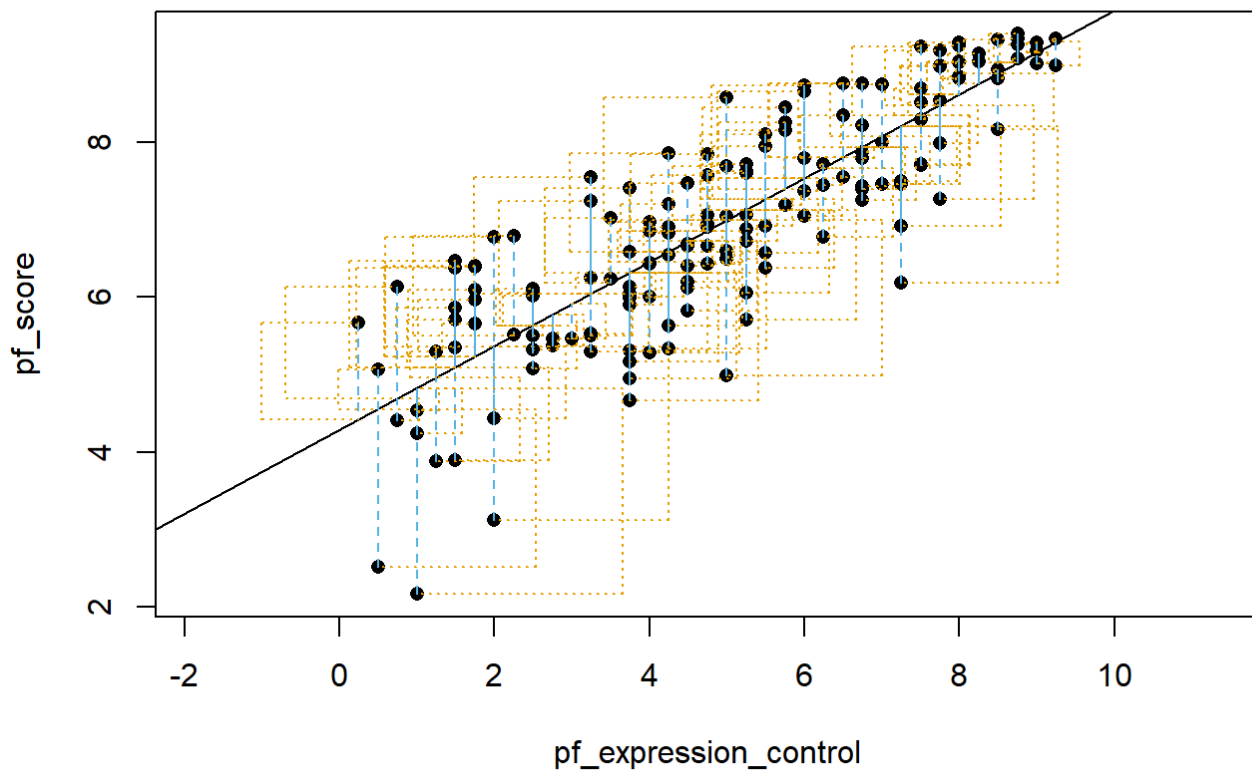
The coff is 0.84

Exercise 4

Looking at your plot from the previous exercise, describe the relationship between these two variables. Make sure to discuss the form, direction, and strength of the relationship as well as any unusual observations.

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```
plot_ss(x = pf_expression_control, y = pf_score, data = hfi_2016, showSquares  
= TRUE)
```

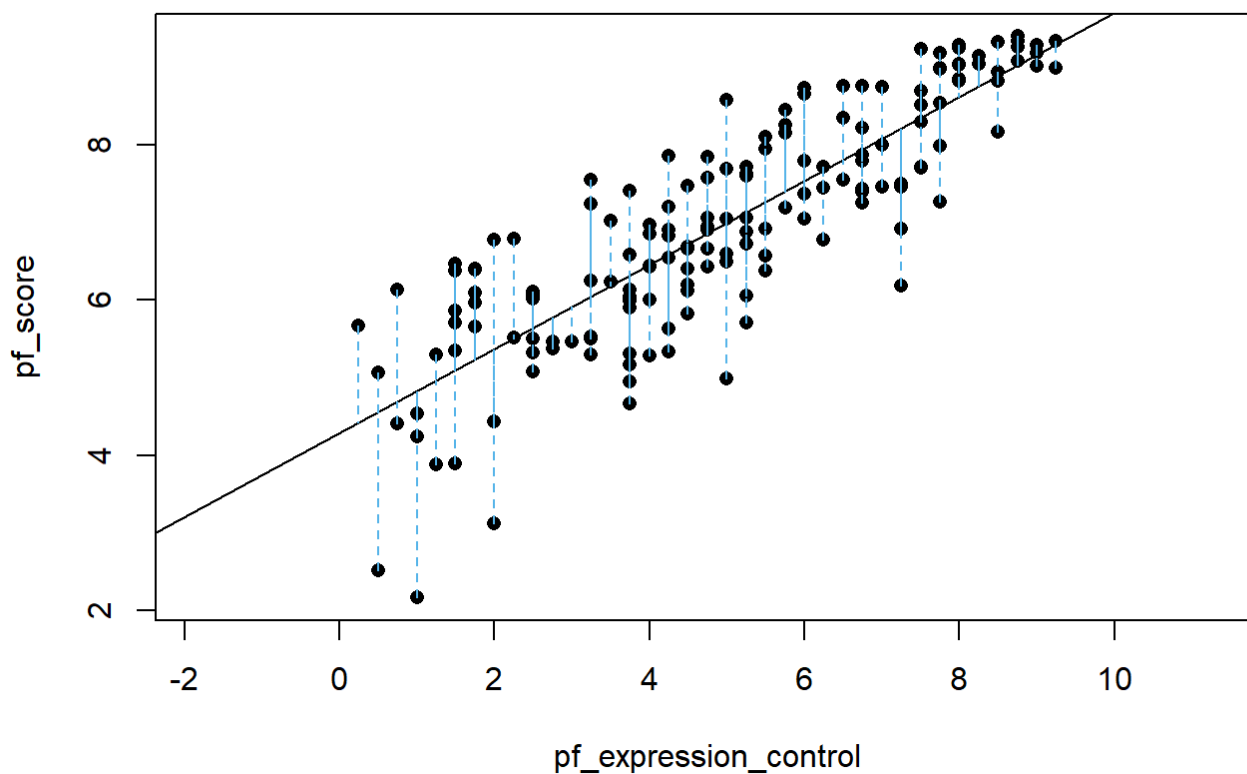


```
## Click two points to make a line.  
  
## Call:  
## lm(formula = y ~ x, data = pts)  
##  
## Coefficients:  
## (Intercept)          x  
##      4.2838      0.5418  
##  
## Sum of Squares: 102.213
```

The correlation is positive and strong. As the pf control expression increases the pf score increases as well.

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```
plot_ss(x = pf_expression_control, y = pf_score, data = hfi_2016)
```



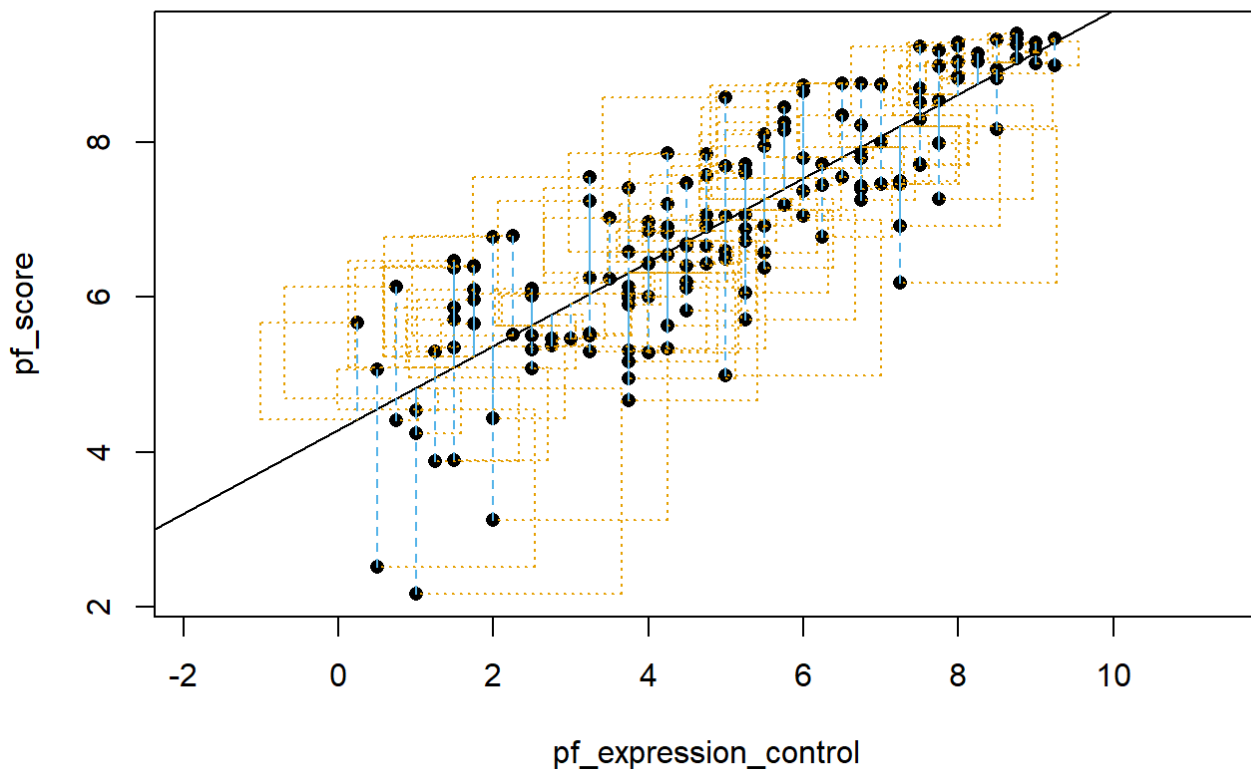
```
## Click two points to make a line.  
  
## Call:  
## lm(formula = y ~ x, data = pts)  
##  
## Coefficients:  
## (Intercept)          x  
##      4.2838      0.5418  
##  
## Sum of Squares: 102.213
```

Exercise 5

Using plotss, choose a line that does a good job of minimizing the sum of squares. Run the function several times. What was the smallest sum of squares that you got? How does it compare to your neighbours?

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```
plot_ss(x = pf_expression_control, y = pf_score, data = hfi_2016, showSquares = TRUE)
```



```
## Click two points to make a line.

## Call:
## lm(formula = y ~ x, data = pts)
##
## Coefficients:
## (Intercept)          x
##      4.2838      0.5418
##
## Sum of Squares: 102.213
```

The smallest sum of Squares was 102.213

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```
m1 <- lm(pf_score ~ pf_expression_control, data = hfi_2016)
tidy(m1)
```

```
## # A tibble: 2 × 5
##   term                estimate std.error statistic  p.value
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)        4.28      0.149      28.8 4.23e-65
## 2 pf_expression_control 0.542    0.0271     20.0 2.31e-45
```

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```
glance(m1)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##   <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
## 1   0.714      0.712 0.799      400. 2.31e-45     1 -193.  391.  400.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Exercise 6

Fit a new model that uses `pf_expression_control` to predict `hf_score`, or the total human freedom score. Using the estimates from the R output, write the equation of the regression line. What does the slope tell us in the context of the relationship between human freedom and the amount of political pressure on media content?

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```
lm1 <- lm(pf_score ~ pf_expression_control, data = hfi_2016)

tidy(lm1)
```

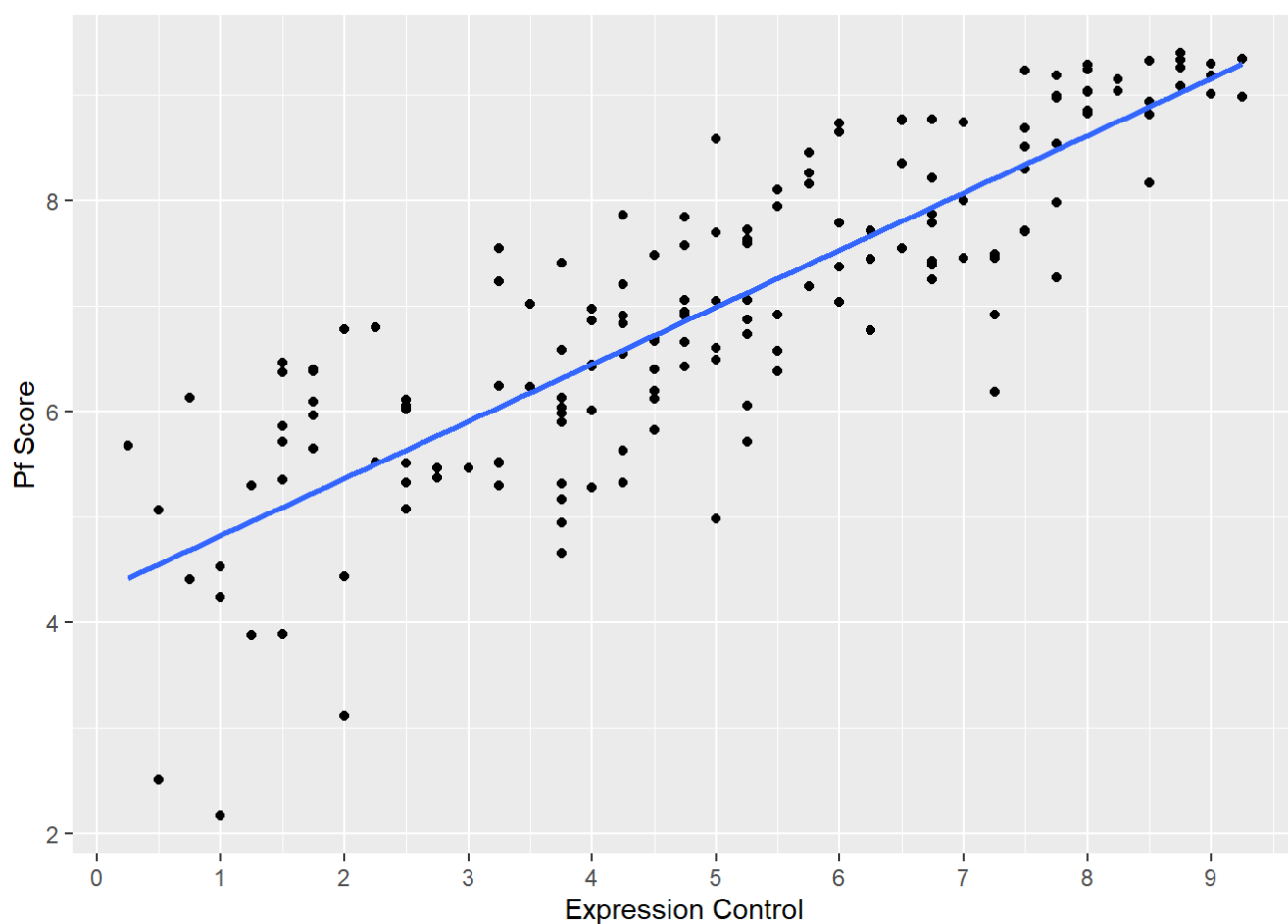


```
## # A tibble: 2 × 5
##   term                estimate std.error statistic  p.value
##   <chr>                <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)          4.28      0.149     28.8 4.23e-65
## 2 pf_expression_control 0.542    0.0271    20.0 2.31e-45
```

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```
hfi_2016 %>%
  ggplot(aes(x = pf_expression_control, y = pf_score)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "Expression Control",
       y = "Pf Score") +
  theme(plot.title = element_text(hjust = 0.50)) +
  scale_x_continuous(breaks = seq(0, 10, 1))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


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```
table22 <- lm(hf_score ~ pf_expression_control, data = hfi_2016)
summary(table22)
```

```
##
## Call:
## lm(formula = hf_score ~ pf_expression_control, data = hfi_2016)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.68164 -0.45467  0.05692  0.46699  1.88128
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.05340    0.12293   41.11  <2e-16 ***
## pf_expression_control  0.36843    0.02236   16.48  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6595 on 160 degrees of freedom
## Multiple R-squared:  0.6291, Adjusted R-squared:  0.6268
## F-statistic: 271.4 on 1 and 160 DF,  p-value: < 2.2e-16
```

Exercise 7

If someone saw the least squares regression line and not the actual data, how would they predict a country's personal freedom school for one with a 3 rating for pf_expression_control? Is this an overestimate or an underestimate, and by how much? In other words, what is the residual for this prediction?

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```
pf_exp_control <- 3

pf_score6 <- 4.28 + 0.542 * pf_exp_control
pf_score6
```

```
## [1] 5.906
```

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```
hfi_2016 %>%
  group_by(pf_score) %>%
  filter(pf_expression_control == 3.0)
```

```
## # A tibble: 1 × 4
## # Groups:   pf_score [1]
##   year hf_score pf_score pf_expression_control
##   <dbl>   <dbl>   <dbl>           <dbl>
## 1  2016     5.29     5.47             3
```

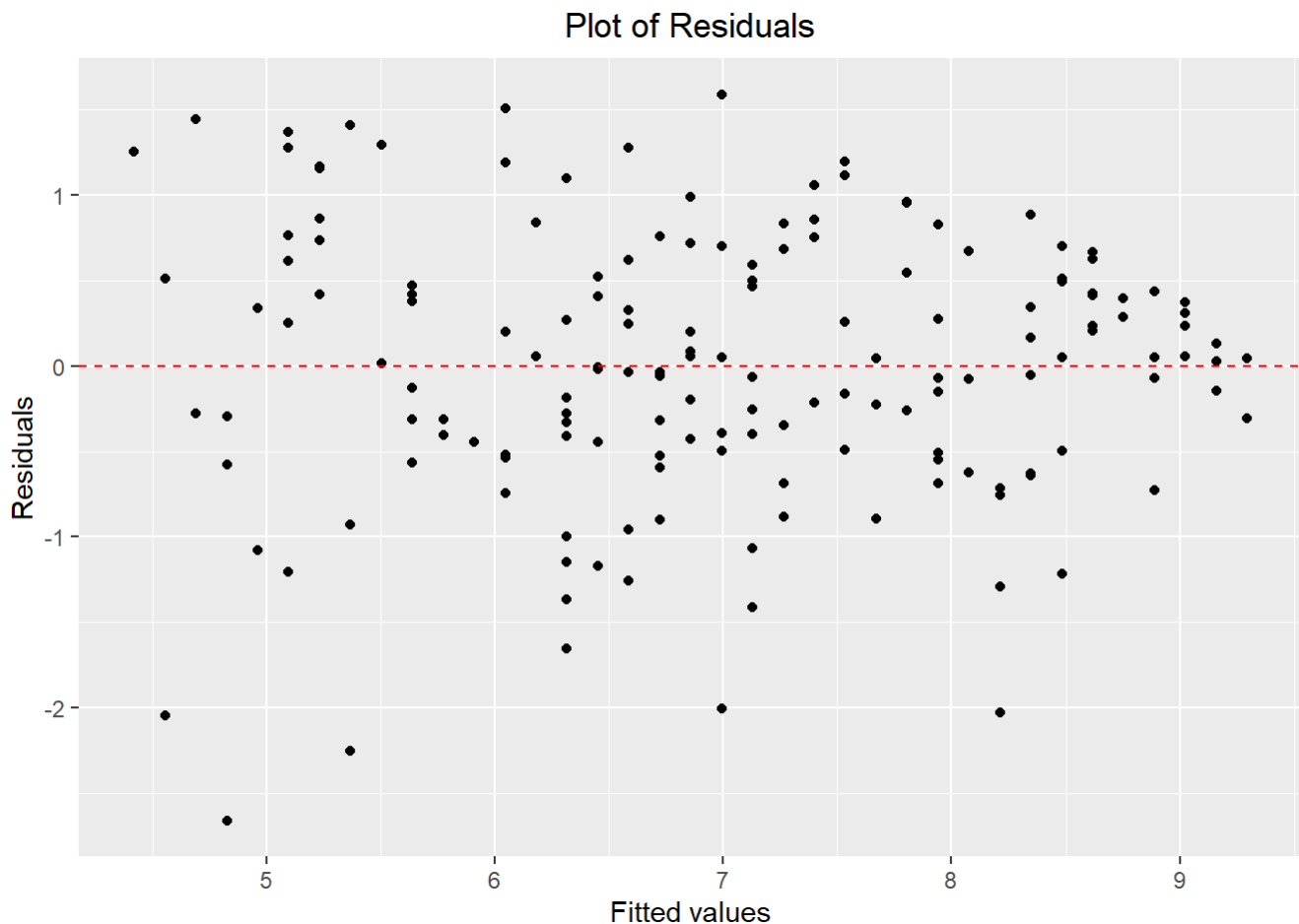
Exercise 8

Is there any apparent pattern in the residuals plot? What does this indicate about the linearity of the relationship between the two variables?

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```
lm1_aug <- augment(lm1)

ggplot(data = lm1_aug, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(
    x = "Fitted values",
    y = "Residuals",
    title = "Plot of Residuals") +
  theme(plot.title = element_text(hjust = 0.50))
```



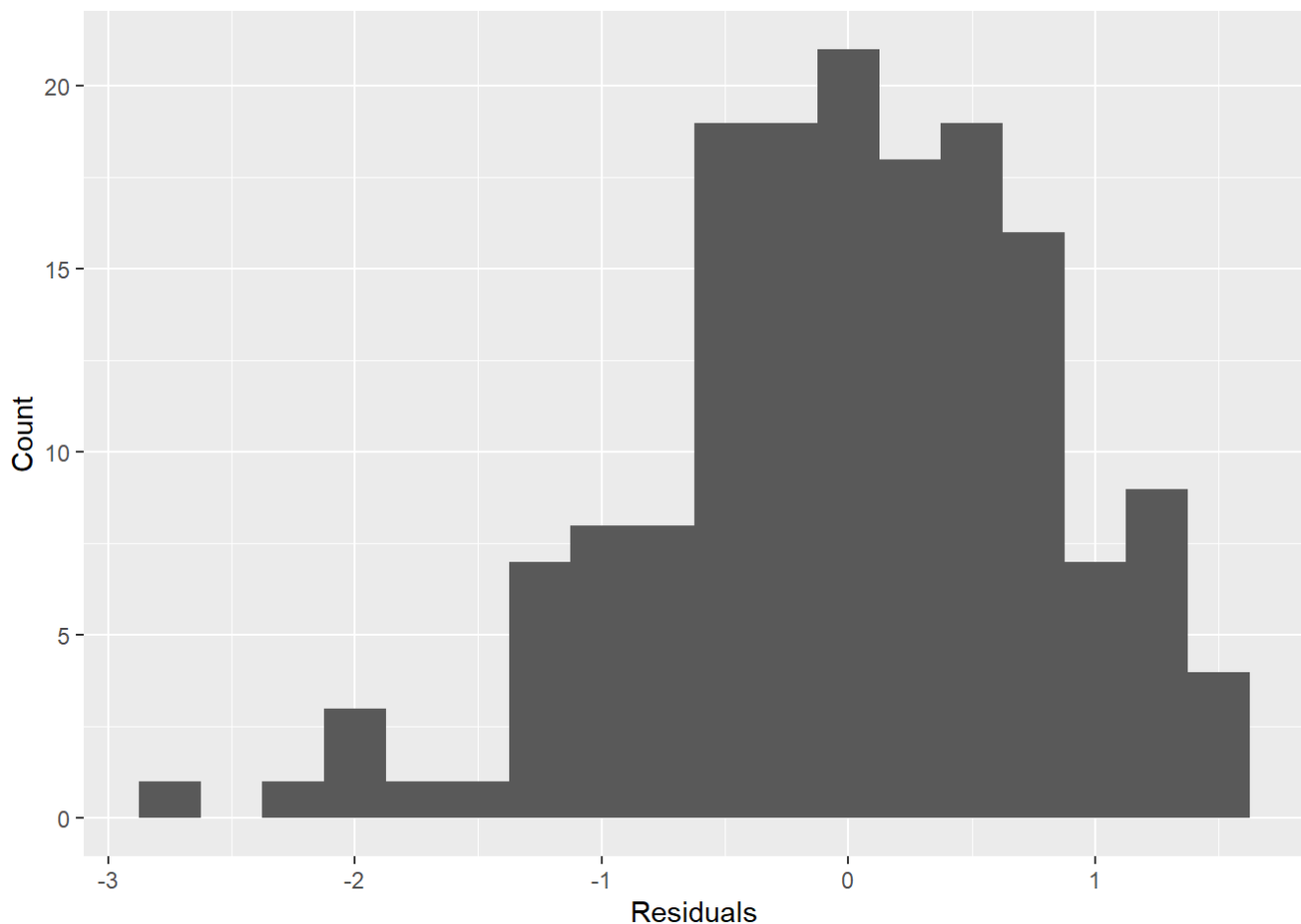
There is no apparent pattern in the residual plot. This indicates that there is a linear relationship between the two variables.

Exercise 9

Based on the histogram, does the nearly normal residuals condition appear to be violated? Why or why not?

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```
ggplot(data = lm1_aug, aes(x = .resid)) +  
  geom_histogram(binwidth = 0.25) +  
  labs(  
    x = "Residuals",  
    y = "Count") +  
  theme(plot.title = element_text(hjust = 0.50))
```

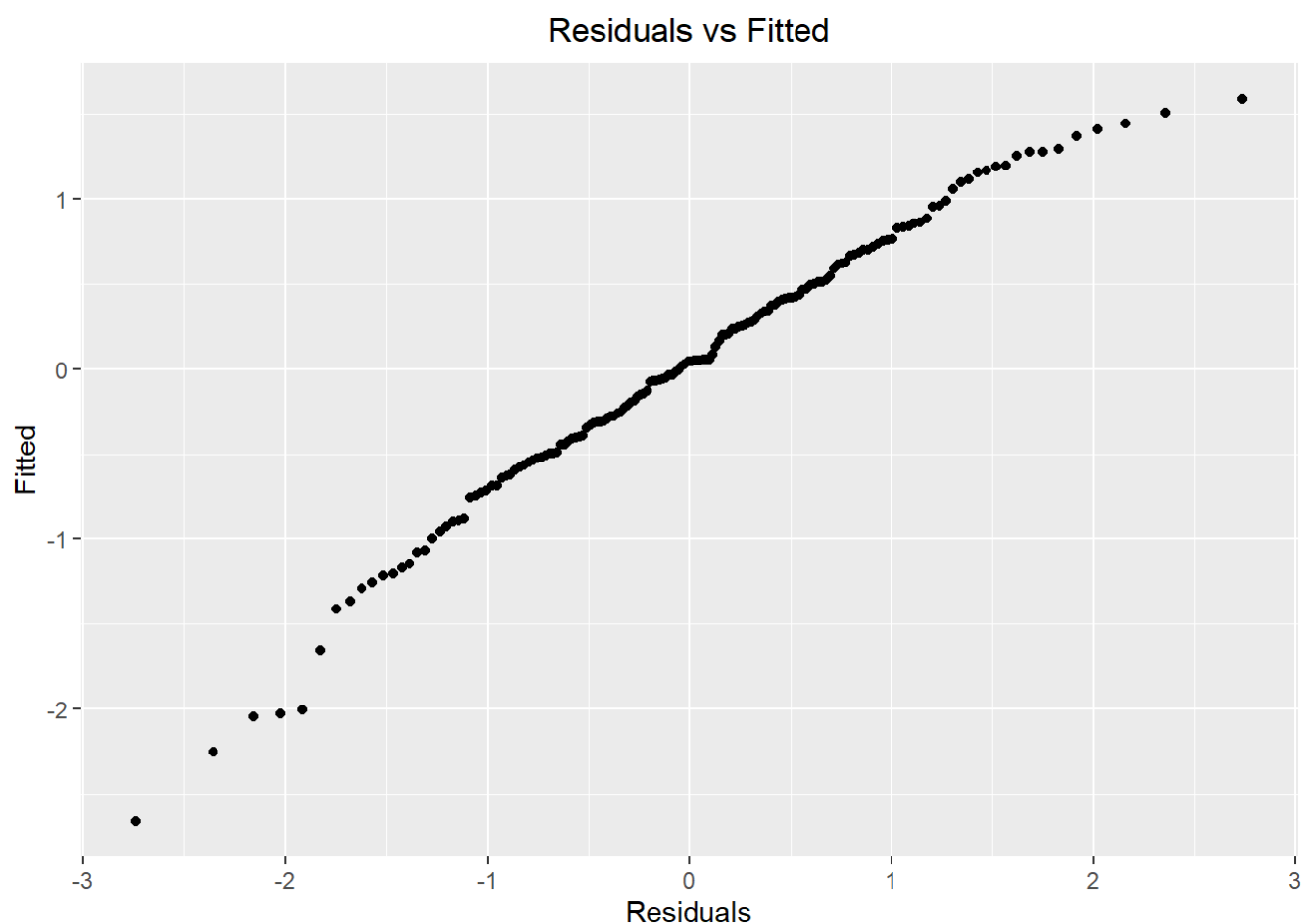


Although the histogram is right skewed, they seem to be fairly distributed. The condition of normal residual is met.

Exercise 10

Based on the residuals vs. fitted plot, does the constant variability condition appear to be violated? Why or why not?

```
ggplot(data = lm1, aes(sample = .resid)) +  
  stat_qq() +  
  labs(  
    x = "Residuals",  
    y = "Fitted",  
    title = "Residuals vs Fitted") +  
  theme(plot.title = element_text(hjust = 0.50))
```



The residuals vs fitted plot do not seem to be violating constant variability. They are scattered and the condition is met.